

Question

1. Dividing a program into functions

- a) is the key to object-oriented programming.
- b) makes the program easier to conceptualize.
- c) May reduce size of program
- d) Make the program faster.

→ Ans. b) makes the program easier to conceptualize

2. A function name must be followed by parentheses.

3. A function body is delimited by braces.

4. Why is the main() function special?

→ Ans. the main() function is special function because it is the first executed when the program starts.

5. A c++ instruction that tells the computer to do something is called a Statement.

6. Write an example of a normal C++ Comment and an example of an old-fashioned /* comment.

→ Ans. // this is example of a normal comment

/* this is multiline

comment */

7. An expression

a) usually evaluates to a numerical value.

b) indicates the emotional state of the program.

c) always occurs outside a function

d) may be part of a statement.

→ Ans. a) usually evaluates to a numerical value.

8. Specify how many bytes are occupied by the following data types in a 32-bit system:

a) Type int

b) Type long double

c) Type float

d) Type long

→ Ans: a) 4 bytes b) 20 bytes c) 4 bytes d) 4 bytes

9. True or false: A variable of type char can hold the value 301.

→ Ans: False

10. What kind of program elements are the following?

a) 12

b) 'a'

c) 4.28915

d) JungleJim

e) JungleJim()

→ Ans: a) integer constant b) character constant c) float constant d) Variable
e) function Name

11. Write statements that display on the screen

a) the character 'x'

b) the name Jim

c) the number 509

→ Ans: a) cout << 'x'; b) cout << "Jim"; c) cout << 509;

12. True or false: In an assignment statement, the value on the left of the equal sign is always equal to the value on the right.

→ Ans: False

13. Write a statement that displays the variable george in a field 10 characters wide.

→ Ans: cout << setw(10) << george;

14. What header file must you #include with your source file to use cout and cin?

→ Ans: #include <iostream>

15. Write a statement that gets a numerical value from the keyboard and places it in the variable temp.

→ Ans: int temp;
cin >> temp;

16. What header file must you #include with your program to use setw?

→ Ans: #include <iomanip>

17. Two exceptions to the rule that the compiler ignores whitespace are string constant and preprocessor directives.

18. True or false: It's perfectly all right to use variables of different data types in the same arithmetic expression.

→ Ans: True

19. The expression $11 \% 3$ evaluates to 2.

20. An arithmetic assignment operator combines the effect of what two operators?

→ Ans: ~~arithmetic~~ arithmetic operators and assignment operators

21. Write a statement that uses an arithmetic ^{assignment} operator to increase the value of the variable temp by 23. Write the same statement without the arithmetic assignment operator.

Ans: `int temp;`
`temp += 23; // using arithmetic assignment operator`
`temp = temp + 23; // without using assignment operator`

22. The increment operator increases the value of a variable by how much?

Ans: The increment operator increases the value of a variable by 1.

23. Assuming var1 start with the value 20, what will the following code fragment print out?

`cout << var1--;`

`cout << ++var1;`

→ Ans: 2020

24. In the example, we have seen so far, header files have been used for what purpose?

→ Ans: Header files have been used for providing declarations and other data for library functions, overloaded operator and object.

25. The actual code for library function is contained in a library file.